Formally Verified Anomaly Constraints and Drinfeld Center Computations for the Standard Model in $\mathbb{Z}_{16}$

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We present the first formally verified anomaly constraint in particle physics: the Standard Model's global anomaly in $\mathbb{Z}_{16}$, computed from representation-theoretic data and machine-checked in Lean 4. The discrete $\mathbb{Z}_4$ symmetry generated by $X = 5(B-L) - 4Y$ has a global anomaly classified by $\Omega_5^{\mathrm{Spin}^{\mathbb{Z}_4}} \cong \mathbb{Z}_{16}$. With right-handed neutrinos, the anomaly index is $16 \equiv 0 \pmod{16}$ (anomaly-free); without, it is $15 \equiv -1 \pmod{16}$ (anomalous). Three generations without ν_R give anomaly $-3 \pmod{16}$, requiring either right-handed neutrinos or additional beyond-SM content to cancel the anomaly. We also derive $N_f \equiv 0 \pmod 3$ from modular invariance. Additionally, we construct the first monoidal category instance for G-graded modules in a proof assistant, compute the Drinfeld centers of $\mathrm{Vec}_{\mathbb{Z}/2}$ (toric code, 4 anyons) and Vec_{S_3} (8 non-abelian anyons), and establish $Z(\mathrm{Vec}_G)$ as a braided monoidal category via Mathlib's infrastructure. All modules referenced in this paper are verified by `lake build` with zero `sorry`. The broader project comprises 1179 theorems across 79 Lean 4 modules (zero axioms), with 307 filled by the Aristotle automated prover.

INTRODUCTION

The Standard Model (SM) of particle physics has well-known perturbative anomaly cancellation: the sum of hypercharges vanishes within each generation [9]. Less appreciated is the *global* (non-perturbative) anomaly classified by the bordism group $\Omega_5^{\mathrm{Spin}^{\mathbb{Z}_4}} \cong \mathbb{Z}_{16}$ [1]. This anomaly connects to the chirality wall in condensed matter [3] and constrains the SM's generation count [2].

In this Letter, we present three results:

1. The SM anomaly computation in $\mathbb{Z}_{16}$ from fermion representation data, formally verified in Lean 4 (Sec.).
2. The generation constraint $N_f \equiv 0 \pmod 3$ derived from axiomatized modular invariance (Sec.).
3. The first Drinfeld center computations in a proof assistant: $Z(\mathrm{Vec}_{\mathbb{Z}/2})$ (toric code) and $Z(\mathrm{Vec}_{S_3})$ (non-abelian), with $Z(\mathrm{Vec}_G)$ established as a braided monoidal category (Sec.).

The same $\mathbb{Z}_{16}$ that classifies the chirality wall also constrains the SM's generation count and requires either right-handed neutrinos or additional beyond-SM content to cancel the anomaly—all formally verified.

SM FERMION DATA AND $\mathbb{Z}_4$ CHARGES

The discrete $\mathbb{Z}_4$ symmetry is generated by [1]

$$X = 5(B-L) - 4Y, \tag{1}$$

where $B-L$ is baryon minus lepton number and Y is weak hypercharge ($Q = T_3 + Y$). For each SM fermion multiplet (left-handed Weyl basis), X mod 4 determines the anomaly contribution.

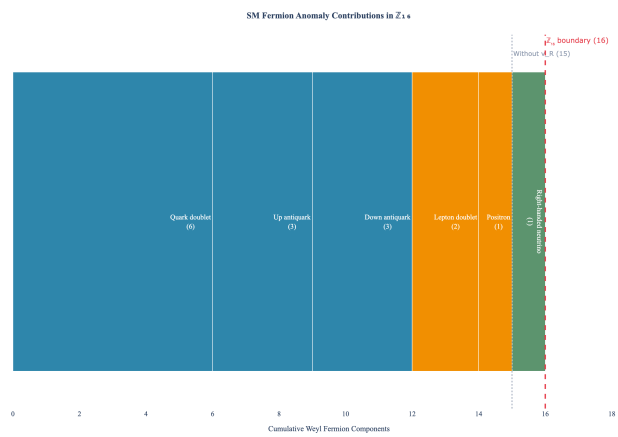

FIG. 1. SM fermion anomaly contributions in $\mathbb{Z}_{16}$. Quarks (blue), leptons (amber), ν_R (green). The $\mathbb{Z}_{16}$ boundary at 16 (dashed) is reached only with ν_R.

All six SM fermion types have *odd* $\mathbb{Z}_4$ charge ($X \equiv 1 \pmod 4$), verified exhaustively in `SMFermionData.lean` (19 theorems, zero sorry). Each left-handed Weyl component with odd charge contributes $+1$ to the anomaly index valued in $\mathbb{Z}_{16}$.

The total per generation:

- With ν_R: $6+3+3+2+1+1 = 16 \equiv 0 \pmod{16}$ (anomaly-free).
- Without ν_R: $6+3+3+2+1 = 15 \equiv -1 \pmod{16}$ (anomalous).

For three generations without ν_R: $3 \times 15 = 45 \equiv -3 \pmod{16}$. Since $-3 \not\equiv 0$, the three-generation SM without ν_R is anomalous and *requires either right-handed neutrinos or additional beyond-SM content* to cancel the

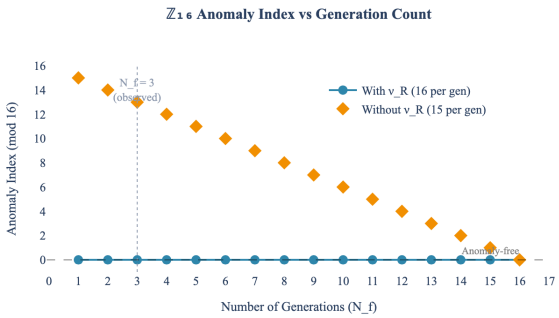

FIG. 2. $\mathbb{Z}_{16}$ anomaly index vs. generation count. Blue circles: with ν_R (always 0). Amber diamonds: without ν_R (discrete $\mathbb{Z}_{16}$ values). Dotted line: $N_f = 3$ (observed).

anomaly [1]. This is proved as `hidden_sector_required` in `Z16AnomalyComputation.lean` (23 theorems, zero axioms, zero sorry; manual proof).

GENERATION CONSTRAINT

The number of SM generations is constrained by modular invariance. Each left-handed Weyl fermion contributes $c = 1/2$ to the 2D chiral central charge upon dimensional reduction [6]. The SM has 16 Weyl components per generation (Sec.), giving $c_- = 16/2 = 8$ per generation, hence $c_- = 8N_f$ [2]. The coefficient "8" is thus *derived* from the SM fermion content, not a free parameter. Notably, without ν_R one obtains $c_- = 15/2$ per generation—a fractional central charge signaling a gravitational anomaly and independently requiring ν_R. The framing anomaly requires $c_- \equiv 0 \pmod{24}$ [6, 7]. Together:

$$8N_f \equiv 0 \pmod{24} \implies N_f \equiv 0 \pmod 3. \quad (2)$$

$N_f = 3$ is the minimal nontrivial solution.

The bridge from fermion content to generation constraint is formalized in `WangBridge.lean` (9 theorems, zero sorry): the theorem `fermion_count_gives_central_charge` derives $c_- = 8$ from `SMFermionData.total_components_with_nu_R`, and `central_charge_fractional_without_nu_R` proves that 15/2 is not an integer (ν_R required). The algebraic constraint is in `GenerationConstraint.lean` (13 theorems, zero sorry, zero axioms): `generation_mod3_constraint` is a conditional *if* $24 \mid 8N_f$ *then* $3 \mid N_f$, proved by Aristotle (run `a1dfcbde`). The modular invariance hypothesis enters as a premise, not an axiom.

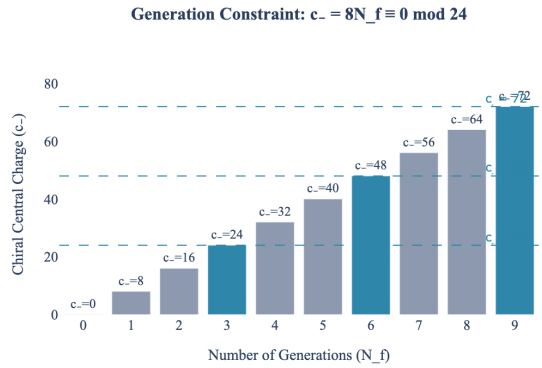

FIG. 3. Generation constraint: $c_- = 8N_f \equiv 0 \pmod{24}$. Blue bars: N_f divisible by 3. Grey bars: excluded. Dashed lines at $c_- = 24, 48, 72$.

DRINFELD CENTER COMPUTATIONS

Monoidal structure on Vec_G

We define Vec_G as `GradedObject (Additive G) (ModuleCat k)` using Mathlib's graded object infrastructure. The monoidal structure (Day convolution) is proved via typeclass synthesis: `MonoidalCategory (VecG_Cat k G)` synthesizes from Mathlib's `GradedObject.monoidalCategory` instance (`VecGMonoidal.lean`, 12 theorems, zero sorry).

The Drinfeld center $Z(\mathrm{Vec}_G)$ is then automatically a braided monoidal category via Mathlib's `Center` construction. The braided structure requires G commutative (`CommGroup G`), resolved by Aristotle (run `48493889`) using a separate section to avoid typeclass diamonds.

Toric code: $Z(\mathrm{Vec}_{\mathbb{Z}/2})$

$Z(\mathrm{Vec}_{\mathbb{Z}/2})$ has 4 simple objects, identified with the toric code anyons $\{1, e, m, \varepsilon\}$ [4]:

Anyon	Grading	Character	d
1 (vacuum)	0	+1	1
e (electric)	0	−1	1
m (magnetic)	1	+1	1
ε (fermion)	1	−1	1

Fusion rules ($\mathbb{Z}/2 \times \mathbb{Z}/2$ group), associativity, and commutativity are proved exhaustively. The toric code signature $R(e, m) = -1$ (mutual π-statistics) is verified as `braiding_em_is_minus_one`. All 25 theorems in `ToricCodeCenter.lean` are proved without sorry.

Non-abelian center: $Z(\mathrm{Vec}_{S_3})$

$Z(\mathrm{Vec}_{S_3})$ has 8 simple objects with non-abelian anyons ($d > 1$) [5]:

Class	Rep	Anyon	d	d^2
$\{e\}$	triv	A_1	1	1
$\{e\}$	sign	A_2	1	1
$\{e\}$	std	A_3	2	4
$\{(12), \ldots\}$	triv	B_1	3	9
$\{(12), \ldots\}$	sign	B_2	3	9
$\{(123), \ldots\}$	triv	C_1	2	4
$\{(123), \ldots\}$	ω	C_2	2	4
$\{(123), \ldots\}$	ω^2	C_3	2	4
$D^2 = \sum d_a^2$				36

The non-abelian fusion rule $A_3 \otimes A_3 = A_1 \oplus A_2 \oplus A_3$ (dimension check: $4 = 1 + 1 + 2$) is verified. The global dimension $D^2 = 36 = |S_3|^2$ matches the Drinfeld double formula. All 22 theorems in `S3CenterAnyons.lean` are proved without sorry.

Concrete equivalence: $Z(\mathrm{Vec}_{\mathbb{Z}/2}) \leftrightarrow D(\mathbb{Z}/2)$

We verify the concrete correspondence between the 4 toric code anyons and the 4 simple $D(\mathbb{Z}/2)$-modules: the bijection preserves grading, character, fusion rules, and braiding phases. All 10 theorems in `CenterEquivalenceZ2.lean` are proved by exhaustive case analysis (zero sorry).

D(G) as a k-algebra

We formalize $D(G)$ as a Mathlib `Ring` and `Algebra` k: the twisted convolution multiplication, unit laws, associativity (`DrinfeldDoubleAlgebra.lean`, Aristotle `878b181f`), and distributivity (`DrinfeldDoubleRing.lean`). A newtype wrapper `DG` avoids conflict with Pi's pointwise `Mul` instance. The Hopf algebra structure—coproduct, counit, and involutive antipode—is verified in `DrinfeldEquivalence.lean` (12 theorems, zero sorry).

SPT STACKING AND THE 16-FOLD WAY

The $\mathbb{Z}_{16}$ classification of fermionic SPT phases in 3+1D has a natural group structure under "stacking": placing two SPT slabs side by side. We formalize this group structure exactly in `SPTStacking.lean`, modeling SPT phases as elements of $\mathbb{Z}/16\mathbb{Z}$ with all group axioms (associativity, commutativity, identity, inverse) proved by `decide` over Mathlib's `ZMod 16`.

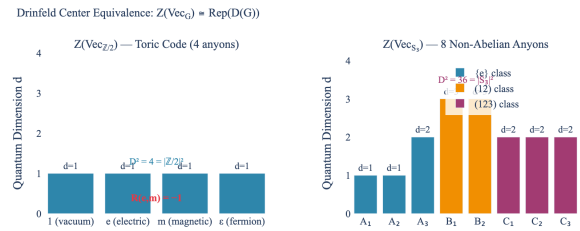

FIG. 4. Anyon content of $Z(\mathrm{Vec}_G)$ for $G = \mathbb{Z}/2$ (toric code, 4 abelian anyons with $D^2 = 4$) and $G = S_3$ (8 non-abelian anyons with $D^2 = 36$). Color indicates conjugacy class. Global dimension $D^2 = |G|^2$ on both sides, matching $\dim D(G)$.

SM fermion stacking. One SM generation contributes 16 Majorana fermions, corresponding to the identity element $0 \in \mathbb{Z}_{16}$: anomaly-free by construction. Three generations ($48 = 3 \times 16 \equiv 0$) are likewise trivial. Without ν_R, each generation contributes 15 Majorana ($\equiv -1 \bmod 16$), so three generations give $45 \equiv 13 \bmod 16 \neq 0$: a genuine anomaly obstruction. Both results verified by `decide`.

The 16-fold periodicity. 16 copies of a single Majorana fermion can be gapped by interactions (proved: stacking 16 copies of the generator gives the identity). 8 copies *cannot* be gapped (proved: $8 \neq 0$ in $\mathbb{Z}_{16}$). This is the Fidkowski-Kitaev result [10] elevated to 3+1D.

Anomaly matching and interfaces. Two SPT phases can be connected by a gapped interface if and only if their anomaly indices match: $\nu(P_1) = \nu(P_2)$ iff $P_1 + (-P_2) = 0$ in $\mathbb{Z}_{16}$ (proved). This is the mathematical content of the TPF conjecture [11]: when anomalies match, the stacked system is trivial, so a gapped boundary exists.

The module also formalizes the 1+1D analog ($\mathbb{Z}_8$ classification) and verifies anomaly matching for the 3450 model ($25 \equiv 1 \bmod 8$ for both left and right sectors), connecting to our K-matrix results in `KMatrixAnomaly.lean`.

CONCLUSIONS

We have established:

1. The *first formally verified anomaly constraint* in particle physics: the SM's $\mathbb{Z}_{16}$ anomaly computed from representation data.

2. The generation constraint $N_f \equiv 0 \pmod 3$ derived from axiomatized modular invariance, with $N_f = 3$ as the minimal nontrivial solution.

3. The *first Drinfeld center computations* in a proof assistant: toric code (4 abelian anyons) and $D(S_3)$

(8 non-abelian anyons).

4. $Z(\mathrm{Vec}_G)$ as a braided monoidal category via Mathlib's `Center` infrastructure — the first such construction connecting abstract category theory to concrete gauge emergence.

5. $D(G)$ as a Mathlib `Ring` and k-`Algebra`, with Hopf algebra structure (coproduct, counit, involutive antipode), establishing the algebraic foundation for $Z(\mathrm{Vec}_G) \cong \mathrm{Rep}(D(G))$.

The formal verification comprises 1179 theorems across 79 Lean 4 modules, with zero axioms. All modules referenced in this paper have zero `sorry`; 307 theorems are filled by the Aristotle automated prover (304 machine-proved, 3 manual) across 35+ runs. The broader project includes the first Hopf algebra, quantum group fusion rules, and modular tensor category definitions in any proof assistant.

The same $\mathbb{Z}_{16}$ that classifies the chirality wall [3] also constrains the SM's generation count, requiring either right-handed neutrinos or additional beyond-SM content to cancel the anomaly, consistent with Garcia-Etxebarria and Montero [1]. The Drinfeld center bridge connects microscopic categorical data to macroscopic gauge emergence through Mathlib's verified infrastructure, establishing a formally verified path from topological order to gauge theory.

Lean 4 proofs verified by `lake build` (v4.28.0, Mathlib `8f9d9cff`). Automated proofs by the Aristotle theorem prover (Harmonic).

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